Code No: 155AA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
ADVANCED COMPUTER ARCHITECTURE
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What is a parallel computer?	[2]
b)	List the advantages of parallelism to improve performance.	[3]
c)	Why do we need virtual memory in computer architecture?	[2]
d)	List the advantages of virtual memory.	[3]
e)	State the need for a cache memory.	[2]
f)	Draw the Instruction Pipeline design.	[3]
g)	What is compound vector processing?	[2]
h)	List the three Generations of Multicomputers.	[3]
i)	Write the disadvantages of multithreading.	[2]
j)	Define fine-grained and coarse-grained parallelism.	[3]

PART – B

(50 Marks)

2.a)	Outline the classification of parallel computer models.	
b)	Discuss about the PRAM model.	[5+5]
	OR	
3.a)	Describe the program flow mechanisms in detail.	
b)	What are the conditions of parallelism? Explain.	[5+5]
4.a)	Outline the Memory Hierarchy with a neat diagram.	
b)	Discuss the applications of parallel processing.	[5+5]
	OR	
5.	Explain the need of superscalar processors and vector processors.	[10]
6.	Elaborate on the Non-Linear Pipeline Processors.	[10]
	OR	
7.	Describe the Sequential and weak consistency models.	[10]

8. Explain the cache coherence and synchronization mechanism. [10]

OR

9. Explain in detail about Message –passing mechanisms. [10]

10.a) Elaborate on the Dataflow and hybrid Architecture.

b) What is Multithreading? Explain the priciples of Multithreading. [5+5]

OR

11. Discuss about Latency-hiding techniques. [10]

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